Activators and Inhibitors in Pattern Formation*

By James P. Keener

A wide class of diffusion reaction equations with activator-inhibitor reaction dynamics are studied analytically. It is shown that when the inhibitor diffuses much faster than the activator, patterns with large amplitude nonuniformities may exist. It is further shown that with a slightly nonuniform source of activator, the patterns which emerge from the time dependent process have a directional preference. The results given coincide with the numerical results of Gierer and Meinhardt [6].

I. Introduction

The fact that diffusion coupled with a reacting chemical system can lead to nonuniformities of the reactants has generated much research interest among biologists, chemists and mathematicians. This fact seems to go against the standard mathematical folklore that diffusion is a stabilizing or smoothing process. However, it has lead many biologists and chemists to propose that such processes are responsible for pattern formation in biological systems.

Diffusion can never cause instability in a one component system, although it can be the cause of traveling waves or stationary fronts [1–4]. In two component systems, the addition of diffusion can destabilize a stable homogeneous reacting system. In a very nice paper explaining the mechanism for this, Segel and Jackson [5] showed that if one of the components is an "activator" and the other an "inhibitor" in a stable arrangement, then the addition of diffusion will destablize the system provided the inhibitor diffuses faster than the activator. This makes sense intuitively, since if the stable uniform configuration of activator and inhibitor is perturbed slightly, then the effectiveness of the inhibitor in restoring the stable equilibrium is reduced because it diffuses away too quickly.

Exploiting this idea in a different way, Gierer and Meinhardt [6] proposed a number of activator-inhibitor models and studied their solutions numerically. All of their models, with parameter values chosen properly, exhibit stable

Professor James P. Keener, Department of Mathematics, University of Arizona, Tucson, AZ 85721.
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nonuniform distributions or patterns. In addition they found that when the source of activator was slightly nonuniform (but monotone), the pattern which was formed had a specific directional preference. Morphogenetic gradients of this type have an important part in the pattern formation theories of a number of biologists [7–9], and are thought to provide a mechanism by which an organism can acquire directional differentiation.

Problems of reaction and diffusion have been studied in the context of pattern formation by numerous authors [10–13]. Aside from numerical methods, most authors resort to methods of bifurcation theory or singular perturbation theory to extract various qualitative features. In this study we shall exploit the largeness of the inhibitor's diffusion coefficient to deduce the structure of the resulting patterns. Our model equations are motivated by the models of Gierer and Meinhardt, and a number of the patterns found numerically by them will be constructed here. Bifurcation methods do apply to problems of this type [10, 11, 14], but have the unfortunate restriction of yielding only small amplitude disturbances, rather than the large amplitude patterns which are typical in a biological context and are found here. We will indicate how the patterns found here are suggested by (and are actually large amplitude extensions of) bifurcation methods.

The organization of this paper is as follows. In Sec. II we motivate our model equations and specify our basic assumptions. In Sec. III we construct the steady state patterns and give algebraic conditions for their existence. In Sec. IV we show that the results of Sec. III continue to hold when saturation effects are included in the model equations. In Sec. V we discuss the evolution of these patterns and explore the implications of nonuniform sources. In Sec. VI we indicate how the solutions of Sec. III are suggested by bifurcation theory. Finally, in Sec. VII we illustrate the results of Sec. III by applying them to a specific model studied by Gierer and Meinhardt.

There is no particular reason to limit the interpretation of these results to molecular biology; many other applications of these results are conceivable. For example, these results also apply to problems of predator-prey communities where the prey is the activator and the predator is the inhibitor. The patterns which are formed then correspond to geographical areas of dense and sparse populations.

II. The Model Equations

Suppose that there are two reacting substances distributed throughout a finite one dimensional vessel with impermeable boundaries. The activator $a(x,t)$ is assumed to activate the production of itself and the inhibitor $h(x,t)$ by auto- and cross-catalysis. The inhibitor is assumed to inhibit the production of itself and the activator. The activator is produced by a source substance $s(x)$, and the rate of auto- and cross-catalysis is assumed to be proportional to the amount of source substance. Finally, both the activator and inhibitor are removed by some removal process such as deterioration at fixed rates. Assuming that each substance diffuses with respective diffusion coefficients D_a and D_h, we have the rate equations

$$\left.\begin{aligned}\frac{\partial a}{\partial t}&=s+sf(a,h)-\mu a+D_a\frac{\partial^2 a}{\partial x^2},\\ \frac{\partial h}{\partial t}&=sg(a,h)-\nu h+D_h\frac{\partial^2 h}{\partial x^2},\end{aligned}\right\}\quad 0<x<1, \tag{2.1a}$$

$$\frac{\partial a}{\partial x}(0,t)=\frac{\partial a}{\partial x}(1,t)=\frac{\partial h}{\partial x}(0,t)=\frac{\partial h}{\partial x}(1,t)=0,$$

where the functions $f(a,h)$, and $g(a,h)$ are assumed to be smooth and satisfy

$$f(a,h)>0,\quad \frac{\partial f}{\partial a}(a,h)>0,\quad \frac{\partial^2 f}{\partial a^2}(a,h)>0,\quad \frac{\partial f}{\partial h}(a,h)<0 \quad \text{for}\quad a>0,\quad h>0, \tag{2.1b}$$

$$g(a,h)>0,\quad \frac{\partial g}{\partial a}(a,h)>0,\quad \frac{\partial g}{\partial h}(a,h)<0 \quad \text{for}\quad a>0,\quad h>0, \tag{2.1c}$$

$$\lim_{h\to\infty} f(a,h)=0,\qquad f(0,h)=0,\qquad f(a,0)>0. \tag{2.1d}$$

The set of equations (2.1) encompasses most of the model equations proposed and studied in [6]. One specific example of these equations is

$$\begin{aligned}\frac{\partial a}{\partial t}&=s+cs\frac{a^2}{h^k}-\mu a+D_a\frac{\partial^2 a}{\partial x^2},\\ \frac{\partial h}{\partial t}&=c's\frac{a^2}{h^l}-\nu h+D_h\frac{\partial^2 h}{\partial x^2},\end{aligned} \tag{2.2}$$

to which all of our results apply and which we shall examine specifically in Sec. VII. Equations (2.1) do not include the effects of saturation of activator production, but these effects can be added with a slight modification of our techniques, and will be discussed in Sec. IV.

The final assumption necessary to set the stage for later sections is that the diffusion coefficient of the inhibitor is much larger than the diffusion coefficient of the activator. Therefore, we assume that $D_h=1/\epsilon$ where $\epsilon\ll 1$ and that all other coefficients are of order one.

III. Nonuniform steady state solutions

In this section we study the steady state solutions of (2.1) in the limit $\epsilon \ll 1$. We seek solutions $a(x), h(x)$ of

$$D_a \frac{d^2a}{dx^2} + s + sf(a,h) - \mu a = 0, \tag{3.1a}$$

$$\frac{1}{\epsilon}\frac{d^2h}{dx^2} + sg(a,h) - \nu h = 0, \tag{3.1b}$$

$$a'(0) = a'(1) = h'(0) = h'(1) = 0,$$

where $s > 0$ is assumed to be uniform in x and the functions $f(a,h)$, $g(a,h)$ satisfy the properties (2.1b–d).

Since $\epsilon \ll 1$, we assume that there is a power series expansion for $a(x;\epsilon)$ and $h(x;\epsilon)$ of the form

$$\begin{aligned} h(x;\epsilon) &= h_0(x) + \epsilon h_1(x) + \cdots, \\ a(x;\epsilon) &= a_0(x) + \epsilon a_1(x) + \cdots. \end{aligned} \tag{3.2}$$

Multiplying (3.1b) through by ϵ we have

$$\begin{gathered} \frac{d^2h}{dx^2} + \epsilon\left[sg(a,h) - \nu h\right] = 0, \\ h'(0) = h'(1) = 0, \end{gathered} \tag{3.3}$$

which is apparently a regular perturbation problem for $h(x)$ provided $a(x)$ is known. In this paper we will not discuss in detail the convergence properties of the assumed power series solutions; however, at this point it is evident that for known $a(x)$ the power series for $h(x;\epsilon)$ should be convergent for some nonzero radius of convergence ϵ_0. Using (3.2) and expanding (3.3) in powers of ϵ, we find that the first few terms must satisfy

$$\frac{d^2h_0}{dx^2} = 0,\ h_0'(0) = h_0'(1) = 0, \tag{3.4a}$$

$$\frac{d^2h_1}{dx^2} = -\left[sg(a_0,h_0) - \nu h_0\right], \qquad h_1'(0) = h_1'(1) = 0. \tag{3.4b}$$

The solution of (3.4a) is immediately obvious as

$$h_0(x) = H_0 = \text{const.} \tag{3.5a}$$

The solution of (3.4b) is also straightforward (as are all higher order equations), and we find

$$h_1(x) = \int_0^x \int_0^{\xi} \left[\nu H_0 - sg(a_0(\eta), H_0) \right] d\eta \, d\xi + H_1, \tag{3.5b}$$

where the boundary condition $h_1'(1) = 0$ is satisfied if and only if

$$\nu H_0 = \int_0^1 sg(a_0(x), H_0)\, dx. \tag{3.5c}$$

This condition apparently determines the constant H_0. In fact, this condition is a Fredholm alternative in disguise, since the problem

$$u'' = 0, \qquad u'(0) = u'(1) = 0, \tag{3.6a}$$

has the nontrivial solution $u = u_0 \neq 0$. Therefore, the problem

$$u'' = f, \qquad u'(0) = u'(1) = 0, \tag{3.6b}$$

can be solved if and only if

$$\langle u_0, f \rangle = \int_0^1 f(x) u_0 \, dx = u_0 \int_0^1 f(x)\, dx = 0. \tag{3.6c}$$

The condition (3.5c) does indeed determine a unique value for the given $a_0(x)$. For any fixed positive $a_0(x)$, the nonlinear functional

$$G(a_0, H) = \int_0^1 g(a_0(x), H)\, dx \tag{3.7}$$

is positive for $H = 0$ (possibly unbounded) and monotonic nonincreasing in H. Therefore, there is a unique solution H_0 of the equation (3.5c) for each fixed s and $a_0(x)$. Higher order equations for $h_i(x)$, $i = 2, 3, \ldots$, can be solved in a similar way. The equation for $h_i(x)$ is of the form (3.6b), where the right hand side $f_i(x)$ will be completely known except for the constant H_{i-1}, which appears in a linear fashion. The Fredholm alternative can always be satisfied by the proper choice of H_{i-1}, assuming the technical condition

$$\nu - s\int_0^1 g_h(a_0(x), H_0)\, dx \neq 0. \tag{3.8}$$

This condition is always satisfied, since H_0 is an isolated solution of (3.5c) [or alternatively, since $g_h(a, h) \leq 0$], and therefore the perturbation procedure for $h(x;\epsilon)$ given $a(x;\epsilon)$ and $s > 0$ is well defined to all orders of ϵ.

For each positive $a(x)$ there is a solution of (3.1b) of the form $h(x)=H_0+O(\epsilon)$, where H_0 must satisfy $vH_0=sG(a(x),H_0)$, where $G(a,H_0)$ is the functional defined by (3.7). Substituting (3.2) into (3.1a), we find that $a_0(x)$ must satisfy

$$D_a\frac{d^2a_0}{dx^2}+s+sf(a_0,H_0)-\mu a_0=0, \qquad a_0'(0)=a_0'(1)=0, \tag{3.9a}$$

$$vH_0=sG(a_0(x),H_0). \tag{3.9b}$$

This can be viewed as an integrodifferential equation for $a_0(x)$, since H_0, which appears in the differential equation, is defined implicitly by an integral of $a_0(x)$. However, to solve (3.9), we will seek a family of solutions $a_0(x,H_0)$ of (3.9a) parametrized by H_0 and then show that at least one value of H_0 can be picked to satisfy (3.9b).

Equation (3.9a) can be multiplied by da/dx and integrated once to yield

$$\frac{D_a}{2}\left(\frac{da}{dx}\right)^2+sa+sF(a,H_0)-\frac{\mu a^2}{2}=E, \tag{3.10}$$

where $\partial F(a,H_0)/\partial a=f(a,H_0)$ and E is an arbitrary constant. To satisfy the boundary conditions, the trajectory $a(x)$ must connect zeros of the function

$$I(a,H_0)-E=sa+sF(a,H_0)-\frac{\mu a^2}{2}-E. \tag{3.11}$$

Typical plots of the functions

$$\mathcal{F}(a,H_0)=s+sf(a,H_0)-\mu a \tag{3.12}$$

and $I(a,H_0)$ are given in Figs. 1 and 2 respectively for $s>0$ fixed and different values of H_0. Define $H^*(s)$ to be that value of H for which the function $\mathcal{F}(a,H)$

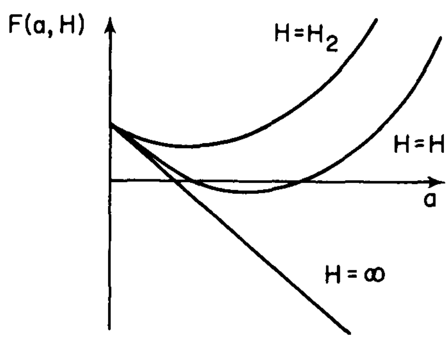

Figure 1. Plot of $\mathcal{F}(a_1H)$ for $s>0$ and $H=\infty$, H_1,H_2 with $H_1>H_2$.

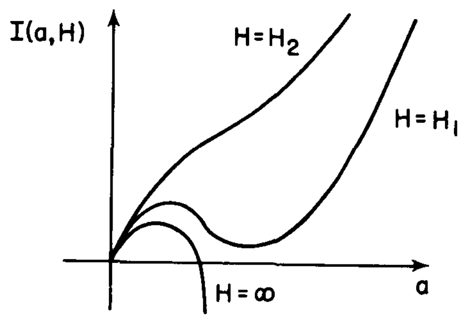

Figure 2. Plot of $I(a,H)$ for $s>0$ and $H=\infty$, H_1, H_2 with $H_1>H_2$.

has a double zero. Then for $H>H^*(s)$ there are two values of a, say $A_-(H)<A_+(H)$ such that $\mathcal{F}(A_\pm(H),H)=0$. For $H<H^*(s)$ the function $\mathcal{F}(a,H)>0$ for $a>0$. An examination of Fig. 2 (or the phase plane plot for (3.9a) in Fig. 3) shows that there are trajectories connecting zeros of $a'(x)$ if and only if $H>H^*(s)$. For $H>H^*(s)$ define E^+ and E^- by

$$E^{\pm}=I(A_{\pm}(H),H). \tag{3.13}$$

The values $E^{\pm}$ are the extremal values of E for which there are admissible trajectories. In the phase plane portrait for Eq. (3.9a) (see Fig. 3) there is a family of closed orbits parametrized by E provided $E^+ \leqslant E \leqslant E^-$. Define $T(E)$ to be the "time" necessary to traverse half of the closed orbit at level E. Since the trajectory for E^- is a separatrix containing a hyperbolic point, $T(E^-)=\infty$. As E approaches E^+, the orbits can be approximated by their linearization about A_+, so that

$$T(E^+)=\lim_{E\to E^+} T(E)=\pi\sqrt{\frac{D_a}{\mathcal{F}_a(A_+(H),H)}}\ . \tag{3.14}$$

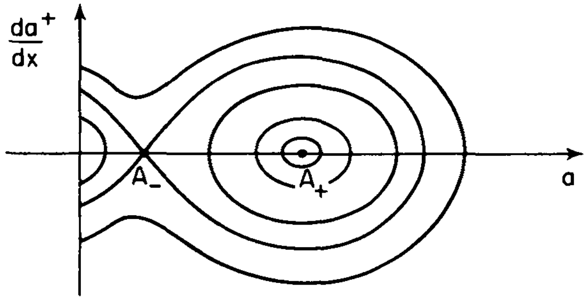

Figure 3. Phase plane portrait for (3.9a) with $H_0>H^*(s)$.

For a given H, suppose that $T(E^+)\leqslant 1$. Then, since $T(E)$ is a continuous function of E, there is a value of of $E=E(H)$ for which $T(E)=1$, and the corresponding trajectory $a(x)=A(x,H)$ satisfies the differential equation (3.9a) and the boundary conditions $a'(0)=a'(1)=0$. Furthermore, $a(x)$ is monotonic.

It remains to show that a specific H can be chosen so that (3.9b) can be satisfied as well. It is apparent that as H approaches $H^*(s)$ from above, the slope $\frac{\partial \mathcal{F}}{\partial a}(A_+(H),H)$ approaches zero, so that $\lim_{H\to H^*(s)} T(E^+)=\infty$. Define $H^{**}(s)$ to be the value of H such that $\infty>H\geqslant H^{**}(s)$ implies $T(E^+(H))\leqslant 1$. To show that (3.9b) has a solution, we will find upper and lower bounds on $G(A(x,H),H)$ and use these bounds to give conditions that guarantee a solution.

A lower bound for $A(x,H)$ is $A_-(H)$. In fact

$$s/\mu<A_-(H)<A(x,H). \tag{3.15}$$

An upper bound for $A(x,H)$ is more difficult to obtain; the idea is to find a region in phase space which contains all of the closed orbits. With the identification $u=a$, $v=a'$, trajectories in phase space are defined by

$$\frac{dv}{du}=-\frac{\mathcal{F}(u,H)}{vD_a} \tag{3.16}$$

Since $f(a,h)$, and hence $\mathcal{F}(u,H)$, is a convex function,

$$\mathcal{F}(a,H)>\mathcal{F}_a(A_-,H)(a-A_-). \tag{3.17a}$$

Therefore, on the line $v=\alpha(u-A_-)$, $u>A_-$,

$$\frac{dv}{du}=-\frac{\mathcal{F}(u,H)}{\alpha(u-A_-)D_a}<-\frac{\mathcal{F}_a(A_-,H)}{\alpha D_a}\leqslant\alpha \tag{3.17b}$$

provided $\alpha\geqslant\sqrt{\mathcal{F}_a(A_-,H)/D_a}$. In other words, in the phase plane, no trajectories crossing the line $v=\alpha(u-A_-)$ can be part of a closed orbit, and all closed orbits lie below the line $v=\alpha(u-A_-)$ and above the line $v=-\alpha(u-A_-)$ with $u>A_-$, where

$$\alpha=\sqrt{-\frac{\mathcal{F}_a(A_-,H)}{D_a}}\ . \tag{3.17c}$$

Because of the convexity of $\mathcal{F}(a,H)$, we also have the inequality

$$\mathcal{F}(a,H)>\mathcal{F}_a(A_+,H)(a-A_+). \tag{3.18a}$$

Applied to the phase plane equation (3.16), this implies that

$$\frac{dv}{du} = -\frac{\mathcal{F}(u,H)}{vD_a} \leqslant -\frac{\mathcal{F}_a(A_+,H)(u-A_+)}{D_a v}, \qquad u > A_+. \tag{3.18b}$$

Therefore, for $u > A_+$ the ellipse

$$v^2 + \frac{\mathcal{F}_a(A_+,H)}{D_a}(u-A_+)^2 = k^2 \tag{3.19a}$$

contains the trajectories of all closed orbits, provided

$$k \geqslant (A_+ - A_-)\sqrt{-\frac{\mathcal{F}_a(A_-,H)}{D_a}}\ . \tag{3.19b}$$

This inequality is derived by requiring an intersection of the ellipse (3.19a) and the line $v = \alpha(u - A_-)$ to occur for $u \geqslant A_+$. Since the ellipse (3.19) contains all closed orbits, the maximal value of u on each orbit must be smaller than the maximal value of u on the ellipse. Therefore, a uniform upper bound for $A(x,H)$ is

$$A(x,H) \leqslant \tilde{A}(H) = A_+ + (A_+ - A_-)\sqrt{-\frac{\mathcal{F}_a(A_-,H)}{\mathcal{F}_a(A_+,H)}}\ . \tag{3.20}$$

The nonlinear function $g(a(x),H)$ is monotonic increasing in a, so that

$$g(s/\mu,H) \leqslant \int_0^1 g(A(x,H),H)\,dx \leqslant g(\tilde{A}(H),H), \qquad H \geqslant H^{**}(s). \tag{3.21}$$

It is clear that we can guarantee the existence of a solution H_0 of (3.9b) by requiring

$$sg(s/\mu, H^{**}(s)) \geqslant \nu H^{**}(s), \tag{3.22a}$$

and

$$sg(\tilde{A}(H),H) \leqslant \nu H \qquad \text{for some} \quad H = \hat{H} \geqslant H^{**}(s). \tag{3.22b}$$

The condition (3.22a) can be replaced by

$$sg(A_-(H),H) > \nu H \qquad \text{for} \quad H = H^{**}(s) \tag{3.22a$'$}$$

by using a more accurate lower bound for $A(x,H)$. The algebraic conditions

(3.22) guarantee that there is a solution of (3.9) with H in the range $H^{**}(s) < H < \hat{H}(s)$. These conditions are generally equivalent to upper and lower bounds on s, in which case (3.22) determines a range of values of s for which the diffusion reaction equations (2.1) have steady state solutions.

PROPOSITION 3.1. *Suppose there are values of $s > 0$ for which* (3.22) *is satisfied. Then for $\epsilon > 0$ sufficiently small, there exist steady state solutions of* (2.1) *for which the function $a(x)$ is monotonic with order one variation in the interval* $0 < x < 1$.

We have not given a complete proof of this proposition. We have only shown the existence of the first terms of the expansion (3.2). It remains to show that higher order terms of this expansion can be found as well, and that the expansion converges for some nonzero ϵ. We prefer to avoid the details of this development and merely invoke the implicit function theorem, which applies, provided the lowest order solution $a_0(x)$ is an isolated solution of (3.1a). The function $a_0(x)$ is isolated in all but exceptional cases, or, in other words, the property that $a_0(x)$ is isolated is generic.

Using the phase plane information about $a_0(x)$ and Eqs. (3.2), (3.5b), we construct in Fig. 4 a typical plot of the distribution of the activator $a(x)$ and inhibitor $h(x)$. It is apparent from (3.5b) and the monotonicity of $g(a,h)$ in a that $h(x)$ must have a shape "similar", although with variation of order ϵ, to the shape of $a(x)$.

It is a simple matter to use the above arguments to deduce the existence of oscillatory structures as well. The proof above depends only on the existence of $H^{**}(s)$ and the conditions (3.22). Recall that $H^{**}(s)$ was defined as that value of H for which $H \geqslant H^{**}(s)$ guarantees that the phase plane half period $T(E^+) \leqslant 1$, where

$$T(E^+) = \pi \sqrt{\frac{D_a}{\mathscr{F}_a(A_+(H),H)}} \quad . \tag{3.14}$$

However, it is clear from (3.14) that we could equally well require $T(E^+) \leqslant 1/n$ for any positive integer n. Then, of course, we would define $H_n^{**}(s)$ so that

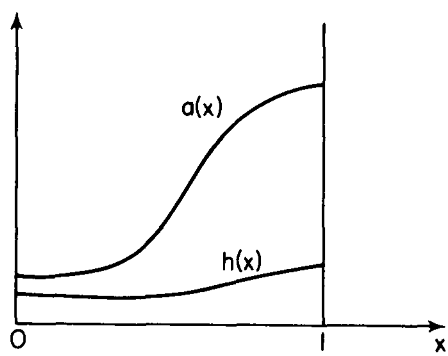

Figure 4. Typical plot of steady state distributions $a(x), h(x)$.

$H \geqslant H^{**}_n(s)$ guarantees $T(E^+) \leqslant 1/n$ and require (3.22) to hold with $H^{**}(s)$ replaced by $H^{**}_n(s)$. If $T(E^+) \leqslant 1/n$, then for each $H \geqslant H^{**}_n(s)$, there is an $E_n(H)$ for which $T(E_n) = 1/n$, and using $n/2$ repetitions of the closed phase plane trajectory for $E = E_n(H)$ gives a periodic solution for $a_0(x)$ with minimal period $2/n$. A typical plot of the resulting steady state solutions of (2.1) is shown in Fig. 5.

There is at least one way to find $H^{**}_n(s)$ which is independent of n. If we assume that at $H = H^{**}_1(s)$, $T(E^+) = 1$, then since $T(E^+)$ is proportional to $\sqrt{D_a}$, we can decrease $T(E^+)$ by simply decreasing D_a. All other estimates and conditions are independent of D_a, so that oscillatory solutions exist for smaller D_a and no other parameter change is required.

PROPOSITION 3.2. *Suppose Proposition* 3.1 *holds for* $D_a = D_1$. *Then*

(a) *Proposition* 3.1 *holds for all* $D_a < D_1$ *as well, and*

(b) *There are oscillatory steady state solutions of* (2.1) *with* $n/2$ *oscillations for all* $D_a \leqslant D_1/n^2$.

These oscillatory solutions are not periodic, because of order ϵ additions to the basic solution $a_0(x)$. This result makes intuitive sense. Decreasing D_a is equivalent to shortening the active range of a perturbation of the activator, so that a disturbance which results from a perturbation in $a(x,t)$ can be expected to have a shorter spatial extent.

We have thus far assumed the existence of $H^{**}(s)$ without further comment. The existence of $H^{**}(s)$ is actually assured by assuming that

$$\pi\sqrt{D_a/\beta} < 1, \tag{3.23a}$$

where

$$\beta = \lim_{H\to\infty} \mathcal{F}_a(A_+(H),H) > 0, \tag{3.23b}$$

because, clearly, if $\pi\sqrt{D_a/\beta} < 1$, then there is a largest value of H for which

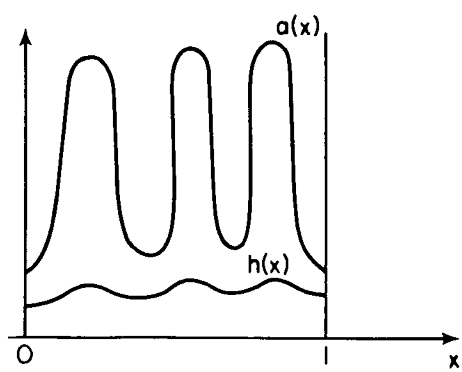

Figure 5. Typical plot of oscillatory steady state distributions $a(x), h(x)$.

$T(E^+)=1$, and this largest value is $H^{**}(s)$. If $f(a,h) \sim c(a^p/h^q)$ for large a and h, then it is straightforward to calculate β. In fact, for $p>1$

$$A_+(H) \sim \left(\frac{\mu}{sc}\right)^{1/(p-1)} H^{q/(p-1)} - \frac{s}{p-1} + \cdots$$

and

$$\mathcal{F}_a(A_+(H),H) \sim (p-1)\mu - s\left(\frac{sc}{\mu H^q}\right)^{1/(p-1)} + \ldots \qquad \text{for large } H,$$

so that $\beta=(p-1)\mu$. Clearly, in this example, $p>1$ guarantees the existence of $H^{**}(s)$ if D_a/μ is small enough.

The algebraic conditions (3.22), (3.23) guarantee the existence of patterns for a (presumably) nontrivial range of values of s for which solutions of (3.1) exist. It is possible to give a simpler condition for these patterns to exist with the handicap that there is no way to determine the range of values of s. Note that $H^{**}(s)$ is a value of H for which

$$T(E^+) = \pi\sqrt{\frac{D_a}{\mathcal{F}_a(A_+(H),H)}} = \frac{1}{n}. \tag{3.24a}$$

At this value of H the solution $A(x,H)$ for $E=E^+$ is the point in phase space $a(x,H)=A_+(H)$. Although this solution is "trivial", it is still a solution of (3.9a). Furthermore, (3.9b) reduces to

$$\nu H = sg(A_+(H),H). \tag{3.24b}$$

If there is a value of $s=s_0$ for which (3.24) has a nonisolated solution, then under appropriate additional conditions (such as the applicability of the implicit function theorem) there is a range of values of s near s_0 for which a solution of (3.9) exists. However, this solution is "small" for s near s_0, and we have no guarantee that the range of s is large enough to give large variation solutions which are of interest. As we will see later, the condition (3.24) is the necessary condition derived by a standard small amplitude bifurcation theory approach to this problem. Although they are often helpful in understanding qualitative features of the solutions, bifurcation conditions do not contain as much information as is contained in (3.22).

A final comment about the conditions (3.22) is in order. These conditions are sufficient but obviously not necessary. The upper and lower bounds for $A(x,H)$ were far too crude to make (3.22) a "best" condition. In fact, it is apparent that the range of values of $A(x,H)$ is an increasing function of the period $T(E)$. Therefore, tighter upper and lower bounds for smaller period solutions could conceivably lead to less restrictive conditions on s.

We can simplify matters somewhat by an additional assumption on $g(a,h)$. If we suppose that the inhibitor production rate saturates in the presence of large amounts of activator, then we can assume that

$$g(a,h) \leqslant K(h), \tag{3.25}$$

where $K(h)$ is a monotonically nonincreasing function of h. With this assumption, (3.22b) reduces to the condition

$$s \geqslant \frac{\nu H}{K(H)} \qquad \text{for some} \quad H \geqslant H^{**}(s).$$

But since $H/K(H)$ is an increasing function of H, we simply require

$$s \geqslant \frac{\nu H^{**}(s)}{K(H^{**}(s))}. \tag{3.26}$$

IV. Alternative conditions on f and g

We can make the model equations more realistic by assuming that the activator production has a saturation level. Suppose that we replace (2.1b) by the assumption

$$f(a,h)>0, \quad \frac{\partial f}{\partial a}(a,h)>0, \quad \frac{\partial^2 f}{\partial a^2}(0,h)>0, \quad \frac{\partial f}{\partial h}(a,h)<0 \tag{4.1}$$

and

$$f(a,h) \leqslant K_2(h) \qquad \text{for} \quad a>0, \quad h>0.$$

With this assumption, the activator is still autocatalytic for small values of a; however, the production rate eventually decreases to zero for large a.

The proof of Propositions 3.1 and 3.2 is only slightly changed. For fixed $s>0$, the function $\mathcal{F}(a,h)$ has at least one and no more than three zeros. [A typical plot of $\mathcal{F}(a,h)$ is shown in Fig. 6.] For h between $\underset{\sim}{H}(s)$ and $\tilde{H}(s)$, $\mathcal{F}(a,h)$ is assumed to have three zeros, $A_1(H)<A_2(H)<A_3(H)$. Furthermore, for h between $\underline{H}(s)$ and $\bar{H}(s)$, with $\underset{\sim}{H}(s<\underline{H}(s)<\bar{H}(s)<\tilde{H}(s)<\infty$, the period $T(E^+) \leqslant 1/n$:

$$T(E^+)=\pi\sqrt{\frac{D_a}{\mathcal{F}_a(A_2(H),H)}} \leqslant \frac{1}{n}. \tag{4.2}$$

As before, there is a family of closed orbits in the phase plane about $A_2(H)$; however, the values of $A(x,H)$ on these orbits are bounded above and below uniformly by

$$\frac{s}{\nu}<A_1(H) \leqslant A(x,H) \leqslant A_3(H). \tag{4.3}$$

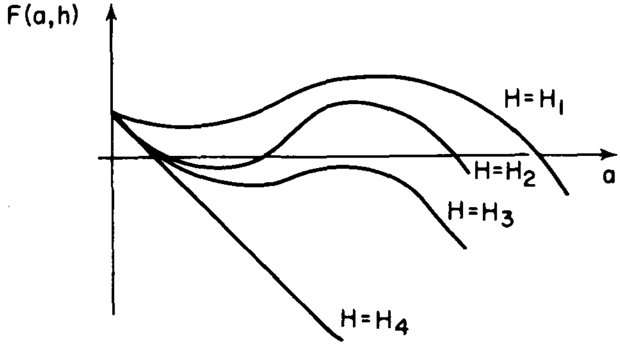

Figure 6. Typical plot of $\mathcal{F}(a,b)$. (See 4.1 for different values of H, $H_1 < H_2 < H_3 < H_4 = \infty$.)

Thus, we can use the monotonicity of $g(a,h)$ in a to deduce conditions similar to (3.22):

$$sg\left(\frac{s}{\nu},\underline{H}(s)\right) \geq \nu\underline{H}(s), \tag{4.4a}$$

$$sg(A_3(H),H) \leq \nu H \qquad \text{for} \quad H=\hat{H}, \qquad \underline{H}(s) \leq \hat{H} \leq \overline{H}(s). \tag{4.4b}$$

An alternative condition for (4.4a) is, naturally

$$sg(A_1(H),H) \geq \nu H \qquad \text{for some } H, \qquad \underline{H}(s) \leq \hat{H} \leq \overline{H}(s). \tag{4.4a$'$}$$

If the condition (4.4) holds for certain values of s, with $\underline{H}(s)$ and $\overline{H}(s)$ defined by (4.2), then nonuniform patterns with $n/2$ oscillations exist as in Propositions 3.1, 3.2.

Finally, if we also assume a saturation level for $g(a,h)$ as in (3.25), then (4.4b) simplifies to

$$s \leq \frac{\nu\underline{H}(s)}{K(\underline{H}(s))}. \tag{4.5}$$

This condition, coupled with (4.4a), depends solely on the value of $\underline{H}(s)$.

In summary, then, Propositions 3.1, 3.2 hold under a varying array of assumptions and conditions, summarized by the following:

Case I (No saturation): If f and g satisfy (2.1b, c, d), then require the conditions (3.22a, b) to hold.

Case II (Saturation of inhibitor production): If f and g satisfy (2.1b, c, d) and g satisfies (3.25), then require the conditions (3.22a) and (3.26) to hold.

Case III (Saturation of activator production): If f and g satisfy (4.1) and (2.1c, d), then require the condition (4.4) to hold.

Case IV (Saturation of activator and inhibitor production): If f and g satisfy (4.1), (2.1c, d) and (3.25), then require the conditions (4.4a) and (4.5) to hold.

V. The initial value problem and the effects of source gradients

In this section we use a combination of standard perturbation techniques and heuristic arguments to examine the time evolution of (2.1) from given initial data. In particular, we will indicate how a nonuniform pattern is formed and why slight gradients in the source $s=s(x)$ lead to a preference in the location of the large activator concentrations.

Consider the equation (2.1) with given initial data

$$\begin{aligned}
&\frac{\partial a}{\partial t}=s+sf(a,h)-\mu a+D_a\frac{\partial^2 a}{\partial x^2},\\
&\frac{\partial h}{\partial t}=sg(a,h)-\nu h+\frac{1}{\epsilon}\frac{\partial^2 h}{\partial x^2},\\
&a_x(0,t)=a_x(1,t)=h_x(0,t)=h_x(1,t)=0,\\
&h(x,0)=\hat{H}_0(x),\qquad a(x,0)=\hat{A}_0(x).
\end{aligned}\tag{2.1}$$

With $\epsilon \ll 1$, the evolution of (2.1) takes place on two different time scales: times scales of order $O(\epsilon)$ and $O(1)$. For times of order ϵ there is an initial layer in which time derivatives of h are of order $1/\epsilon$ and all other derivatives are of order one. To extract the precise behavior we set $\tau=t/\epsilon$ in (2.1) to find

$$\begin{aligned}
&\frac{\partial a}{\partial \tau}=\epsilon\left(s+sf(a,h)-\mu a+D_a\frac{\partial^2 a}{\partial x^2}\right),\\
&\frac{\partial h}{\partial \tau}-\frac{\partial^2 h}{\partial x^2}=\epsilon\left[sg(a,h)-\nu h\right],
\end{aligned}\tag{5.1}$$

with initial data and boundary conditions unchanged from (2.1). Equation (5.1) represents a regular perturbation problem which is readily solvable using a power series in ϵ for $a(x,\tau)$ and $h(x,\tau)$. In particular the first terms of these expansions satisfy

$$\begin{aligned}
&\frac{\partial a_0}{\partial \tau}=0,\qquad \frac{\partial h_0}{\partial \tau}=\frac{\partial^2 h_0}{\partial x^2},\\
&a_0(x,0)=\hat{A}_0(x),\qquad h_0(x,0)=\hat{H}_0(x).
\end{aligned}\tag{5.2}$$

Clearly, $a_0(x,t)=\hat{A}_0=(x)$; and $h_0(x,\tau)$ satisfies the heat equation with no flux boundary conditions and initial data $\tilde{H}_0(x)$ (whose solution is well known) and approaches a uniform concentration for large τ:

$$\lim_{\tau\to\infty} h_0(x,\tau)=\int_0^1 H_0(x)\,dx. \tag{5.3}$$

Therefore, the result in the first "epoch" of time is to smooth out nonuniformities in $\hat{H}_0(x)$.

For times larger than $O(\epsilon)$ we must study the initial value problem

$$\frac{\partial a}{\partial t}=s+sf(a,h)-\mu a+D_a\frac{\partial^2 a}{\partial x^2},$$

$$\frac{\partial^2 h}{\partial x^2}=\epsilon\left[\frac{\partial h}{\partial t}-sg(a,h)+\nu h\right], \tag{5.4}$$

$$a(x,0)=\hat{A}_0(x)+O(\epsilon), \qquad h(x,0)=\int_0^1 H_0(x)\,dx+O(\epsilon),$$

$$a_x(0,t)=a_x(1,t)=h_x(0,t)=h_x(1,t)=0.$$

The exact initial conditions for (5.4) are found by matching with the final values of the initial layer (5.1).

The solution of (5.4) is again a regular perturbation expansion, and we assume that

$$\begin{aligned} a(x,t)&=a_0(x,t)+\epsilon a_1(x,t)+\cdots,\\ h(x,t)&=h_0(x,t)+\epsilon h_1(x,t)+\cdots. \end{aligned} \tag{5.5}$$

Substituting (5.5) into (5.4) we find that

$$\frac{\partial^2 h_0}{\partial x^2}=0, \tag{5.6a}$$

$$\frac{\partial^2 h_1}{\partial x^2}=\frac{\partial h_0}{\partial t}-(sg(a_0,h_0)-\nu h_0), \tag{5.6b}$$

and

$$\frac{\partial a_0}{\partial t}=s+sf(a_0,h_0)-\mu a_0+D_a\frac{\partial^2 a_0}{\partial x^2}. \tag{5.7a}$$

Clearly $h_0(x,t)$ must be independent of x, so that

$$h_0(x,t)=H_0(t).$$

Furthermore, in order to satisfy the boundary conditions for $h_1(x,t)$ [recall the Fredholm alternative for (3.6)], we require that

$$\frac{\partial H_0}{\partial t}=\int_0^1 sg(a_0(x,t),H_0)\,dx-\nu H_0. \tag{5.7b}$$

Equations (5.7a, b) with initial data

$$a_0(x,0)=\hat{A}_0(x), \qquad H_0(0)=\int_0^1 \hat{H}_0(x)\,dx, \tag{5.7c}$$

constitute the system of equations which determines the major effects of pattern formation in this problem.

If the initial data $\hat{A}_0(x)$ and $\hat{H}_0(x)$ are independent of x, the solution $a(x,t)$, $h(x,t)$ is independent of x as well. Then the equations (5.7) reduce to the system of ordinary differential equations

$$\frac{da}{dt}=s+sf(a,H)-\mu a_0=\mathcal{F}(a,H), \tag{5.8a}$$

$$\frac{dH}{dt}=sg(a,H)-\nu H. \tag{5.8b}$$

The solution of (5.8) is somewhat uninteresting in the context of pattern formation. The steady state solution, the solution of

$$\begin{aligned} s+sf(a,H)-\mu a&=0,\\ sg(a,H)-\nu H&=0, \end{aligned} \tag{5.9}$$

is either linearly stable or unstable. [We assume for simplicity that the solution of (5.9) is unique.] If it is stable, then all solutions are attracted to the steady state solution, and if it is unstable, all solutions eventually become unbounded. In neither case do we learn much about pattern formation. If we assume saturation effects as in Sec. 4, then there will probably be more than one solution pair of (5.9); however, this does not alter the arguments that follow.

With nonuniform initial data, the most significant time dependent effects are the result of the structure of $\mathcal{F}(a,h)=s+sf(a,h)-\mu a$. Assume for the moment that H is constant (or nearly constant). For fixed H, the function $\mathcal{F}(a,H)$ has two zeros, A_+ and A_-. The point A_- is a stable zero of (5.8a), and A_+ is an unstable zero. In the absence of diffusion, values of $a<A_+$ will be attracted to A_-, while values of $a>A_+$ will tend to grow without bound. The effect of diffusion is to slow the growth of the growing regions by allowing some of the activator a to migrate to the decreasing regions. If H is held constant for all time, the single component $a(x,t)$ will become unbounded in some regions of space. Stability of the nonuniform steady states requires that H change as well. However, if H is nearly constant initially, the initial growth of $a(x,t)$ is determined with H fixed.

The time dependent behavior is greatly dependent on the initial data. If $a(x,0)$ is initially everywhere smaller (or larger) than $A_+(H)$, then $a(x,t)$ will begin to decrease (or increase) and smooth out, and $H(t)$ will simultaneously evolve much in the way the uniform solution (5.9) evolves. There is no reason to expect a nonuniform pattern from these initial data. Nonuniform patterns will result when portions of $a(x,0)$ are greater than A_+ and other portions are smaller than A_+. Then these sections will initially grow or decrease respectively.

Now, however, H will evolve on a slower time scale, since the effects of the growing and decreasing regions of $a(x,t)$ on $H(t)$ will be neutralized by the integral in (5.7b). In other words, since $g(a,h)$ is monotonic in a, the integral $\int_0^1 sg(a(x,t),H_0)\,dx$ will change more slowly than $a(x,t)$ if a increases in some regions and decreases in others. Thus, if the initial data $\hat{A}_0(x)$ and $\hat{H}_0(s)$ are such that $\partial H_0/\partial t$ in (4.7b) is initially small, then we can expect H_0 to evolve more slowly than $a(x,t)$. As a result $a(x,t)$ will evolve into a nonuniform pattern with H_0 nearly constant, and then this pattern will modulate slightly to bring the two concentrations into steady state balance.

This argument leaves unresolved two contradictions with the numerical calculations of Gierer and Meinhardt. First, the patterns found here have no preference for the location of high concentrations of activator, since the equations (2.1) are symmetric in x. Thus, Fig. 4 could have equally well been drawn with high concentrations on the left. However, Gierer and Meinhardt found a definite preference for the location of high concentrations. Second, Gierer and Meinhardt used uniform initial data for their calculations, and we have predicted either a uniform steady state solution or eventual unbounded solutions.

The resolution of these differences results from small gradients in the source term $s(x)$. Suppose that instead of being constant in x, the function $s(x)$ is monotonic with $s'(x)$ small. The steady state solutions of (2.1) can be found by assuming $s(x)=s_0+\epsilon s_1(x)$. It is not difficult to show, using the obvious perturbation scheme, that a small deviation in $s(x)$ has no major effect on the steady state solutions of the problem.

The major difference comes from the solution of the initial value problem. In fact, since $s(x)$ is x-dependent, the solutions $\mathcal{F}(a,H)=0$ and in particular $A_+=A_+(x)$ will depend on x. The initial dynamics of the activator $a(x,t)$ is determined by its relationship to the value of $A_+(x)$. If the initial datum is larger than A_+ for a given x, it will grow initially, whereas if it is smaller than A_+, it will decrease, thus initiating a favored location for high and low concentrations of $a(x,t)$. Furthermore, if the initial data are uniform and if $a(x,t)=A_+(x)$ for some x, then the nonuniformity of $A_+(x)$ will generate the nonuniformity of $a(x,t)$.

The location of high concentrations depends on the monotonicity properties of $A_+(x)$, since for uniform initial data, decreasing $A_+(x)$ will make $a(x,t)$ more likely to grow. Thus, we can predict the location of high concentrations of $a(x,t)$ for uniform initial data if we know the relationship between $s(x)$ and $A_+(x)$. Clearly, since $\mathcal{F}(a,h)$ is monotonic increasing in s, and $f(a,h)$ and $\mathcal{F}_a(A_+(H),H)$ are positive, A_+ is a monotonic decreasing function of s. Furthermore, nonuniformities in s have no effect on the initial value of H_0 in (5.7). Therefore larger concentrations of $s(x)$ fix the probable locations of large concentrations of activator.

The slight nonuniformity of $s(x)$ has an effect only for uniform or nearly uniform initial data. With uniform $s(x)$, uniform initial data lead to uniform solutions of the initial value problem, and slightly nonuniform initial data can evolve into different nonuniform patterns depending on the shape of the initial nonuniformities. Therefore the uniform data $a(x,0)=A_+(H_0)$ constitute a "separatrix" for the regions of attraction for the nonuniform states. With nonuniform $s(x)$, this "separatrix" is made slightly nonuniform as well, but all stable final patterns are still achievable with appropriate choices of nonuniform initial data. The orientation of $s(x)$ can affect the outcome of the growth only if concentrations are initially nearly uniform. If a concentration has an initial definable structure when the nonuniformity of $s(x)$ is introduced, this nonuniformity can have no major effect on the final pattern.

VI. Application of bifurcation theory

A number of investigators, beginning with Turing [15], have applied bifurcation techniques to reaction-diffusion systems of the form (2.1). In this section we will indicate briefly how the bifurcation theory applies and what it tells us for this problem. We will not carry out the detailed calculations, since they can be found in other places [10, 11, 14].

We begin by seeking a uniform steady state solution of (2.1) which must satisfy

$$s+sf(a,h)-\mu a=0, \tag{6.1a}$$

$$sg(a,h)-\nu h=0. \tag{6.1b}$$

Let A_0, H_0 be a solution pair for (6.1) (which for typical model equations is unique). Then the linear stability of this solution is found by substituting

$$\begin{aligned} a(x,t)&=A_0+\alpha(x,t),\\ h(x,t)&=H_0+\eta(x,t) \end{aligned} \tag{6.2}$$

into (2.1) and linearizing the resulting equations, assuming small α,η. The linearized equations are

$$\begin{aligned} \frac{\partial\alpha}{\partial t}&=(sf_a(A_0,H_0)-\mu)\alpha+sf_h(A_0,H_0)\eta+D_a\frac{\partial^2\alpha}{\partial x^2},\\ \frac{\partial\eta}{\partial t}&=sg_a(A_0,H_0)\alpha+(sg_h(A_0,H_0)-\nu)\eta+D_h\frac{\partial^2\eta}{\partial x^2},\\ &\alpha_x(0,t)=\alpha_x(1,t)=\eta_x(0,t)=\eta_x(1,t)=0. \end{aligned} \tag{6.3}$$

Because of the boundary conditions, we assume a solution of (6.3) of the form

$$\begin{aligned} \alpha&=\alpha_0e^{\sigma t}\cos n\pi x,\\ \eta&=\eta_0e^{\sigma t}\cos n\pi x. \end{aligned} \tag{6.4a}$$

Substituting (6.4a) into (6.3), we find that α_0, η_0 must satisfy the linear homogeneous algebraic equations

$$(M-\sigma I)\begin{pmatrix}\alpha_0\\ \eta_0\end{pmatrix}=\begin{pmatrix} sf_a-\mu-\sigma-n^2\pi^2D_a & sf_h \\ sg_a & sg_h-\nu-\sigma-n^2\pi^2D_h\end{pmatrix}\begin{pmatrix}\alpha_0\\ \eta_0\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}. \tag{6.4b}$$

These equations can have nontrivial solutions if and only if $\det(M-\sigma I)=0$. This enables us to calculate the characteristic values of σ for each n as

$$\sigma=\tfrac{1}{2}\left[\mathrm{Tr}(M)\pm\sqrt{\mathrm{Tr}(M)-4\det(M)}\ \right], \tag{6.5a}$$

where

$$\mathrm{Tr}(M)=s\left(f_a(A_0,H_0)+g_h(A_0,H_0)\right)-\nu-\mu-n^2\pi^2(D_a+D_h),$$

$$\det(M)=\left[sf_a(A_0,H_0)-\mu-n^2\pi^2D_a\right]\left[sg_h(A_0,H_0)-\nu-n^2\pi^2D_h\right]$$

$$-s^2f_h(A_0,H_0)\,g_a(A_0,H_0). \tag{6.5b}$$

If $\mathrm{Tr}(M)>0$ or if $\det(M)<0$, then the solution (6.4a) is linearly unstable (since $\mathrm{Re}\,\sigma>0$). The only stable situation is with $\mathrm{Tr}(M)<0$ and $\det(M)>0$.

The motivation at this point is that if there are parameter values at which the homogeneous steady state solution A_0, H_0 experiences a change of stability, then for nearby parameter values there should be small amplitude nonuniform solutions with shape similar to (6.4a). It is apparent from (6.5) and (2.1) that $\mathrm{Tr}(M)<0$ when $D_h=1/\epsilon$ is large. Therefore, the only possible change of stability is due to a change of sign of $\det(M)$. In the limit $\epsilon\to 0$, with $D_h=1/\epsilon$,

$$\mathrm{sgn}(\det(M))=-\mathrm{sgn}\left(sf_a(A_0,H_0)-\mu-n^2\pi^2D_a\right) \tag{6.6a}$$

For a solution (A_0, H_0) for which $sf_a(A_0,H_0)-\mu>0$, there is an exchange of stability whenever

$$sf_a(A_0,H_0)-\mu=n^2\pi^2D_a. \tag{6.6b}$$

This exchange of stability can occur independently of whether the steady state A_0, H_0 is stable to uniform perturbations. The stability of A_0, H_0 to uniform perturbations is determined by the sign of $\mathrm{Re}(\sigma)$ in (6.5a) with $n=0$ in (6.5b). Clearly the size of D_h or D_a has no effect on this stability, and it is certainly possible to have $sf_a(A_0,H_0)-\mu>0$ and A_0, H_0 stable to uniform perturbations.

This linear stability analysis leads us to a number of conclusions. First, if (6.6b) is satisfied for some fixed k, we expect from bifurcation theory that the full nonlinear equations have nontrivial solution of the form $a=A_0+\alpha\cos n\pi x$

with α small for certain neighboring values of the parameters of (6.6). The condition (6.6), (6.1b) is exactly the condition (3.24) derived in Sec. 3 as a condition for the existence of small nonuniform steady state solutions.

Second, by decreasing D_a in (6.6) we can destabilize the uniform steady state to perturbations of the form $\alpha\cos n\pi x$ for increasingly large n. This is consistent with the observation of Sec. 3 that decreasing D_a introduces nontrivial steady states solutions with $2/n$ oscillations in $0 \leqslant x \leqslant 1$.

VII. A specific example

In this section we illustrate the results of Sec. 3 by applying them to a specific model studied by Gierer and Meinhardt [6]:

$$\frac{\partial a}{\partial t} = s + cs\frac{a^2}{h^k} - \mu a + D_a\frac{\partial^2 a}{\partial x^2},$$

$$\frac{\partial h}{\partial t} = c'\frac{sa^p}{h^l} - \nu h + \frac{1}{\epsilon}\frac{\partial^2 h}{\partial x^2}. \tag{2.2}$$

Clearly for this problem we have

$$f(a,h) = c\frac{a^2}{h^k}, \tag{7.1a}$$

$$g(a,h) = c'\frac{a^p}{h^l}, \tag{7.1b}$$

and

$$\mathcal{F}(a,h) = s + s\frac{ca^2}{h^k} - \mu a, \tag{7.1c}$$

so that the zeros of $\mathcal{F}(a,h)$ are given by

$$A_\pm(h) = \frac{h^k\mu}{2sc}\left[1 \pm \sqrt{1 - \frac{4s^2c}{\mu^2 h^k}}\,\right]. \tag{7.1d}$$

To calculate $H^{**}(s)$, we note that

$$\mathcal{F}_a(a,h) = \frac{2sc}{h^k}a - \mu, \tag{7.2a}$$

so that, evaluated at $A_\pm(h)$,

$$\mathcal{F}_a(A_\pm(H),H) = \pm\sqrt{\mu^2 - \frac{4s^2c}{h^k}}\,. \tag{7.2b}$$

The function $\mathcal{F}_a((A_+(H),H)$ is monotonic increasing in H, so that $T(E^+(H) < 1/n$ whenever

$$H > H^{**}(s) = \left(\frac{4s^2c}{\mu^2 - n^4\pi^4D_a^2}\right)^{1/k} \qquad \text{provided} \quad \mu > \pi^2n^2D_a. \tag{7.2c}$$

If $\mu < \pi^2n^2D_a$, $H^{**}(s)$ does not exist and the corresponding solutions do not exist.

We check the condition (3.22a) using (7.1b) and (7.2c) and find that for $H = H^{**}(s)$,

$$\frac{s}{\nu} g\left(\frac{s}{\nu}, H^{**}(s)\right) > H^{**}(s)$$

provided

$$s^{p+1-2(l+1)/k} > \frac{\nu\mu^p}{c'}\left(\frac{4c}{\mu^2 - n^4\pi^4D_a^2}\right)^{(l+1)/k}. \tag{7.3}$$

This provides a lower bound for s if $k(p+1) > 2(l+1)$ and provides an upper bound otherwise. To check the condition (3.22b) we must first calculate $\tilde{A}(H)$. Using (7.2b), we find that

$$\tilde{A}(H) = 2A_+(H) - A_-(H) = \frac{H\mu}{2sc}\left[1 + 3\sqrt{1 - \frac{4s^2c}{\mu^2H}}\right], \tag{7.4a}$$

which for large H satisfies

$$\tilde{A}(H) \sim \frac{H\mu}{sc}\left[2 - \frac{3s^2c}{\mu^2H} + \cdots\right]. \tag{7.4b}$$

For large H (3.22b) is satisfied if

$$\frac{s}{\nu}\left(\frac{2H\mu}{sc}\right)^p\left(1 - \frac{3p}{2}\frac{s^2c}{\mu^2H} + \cdots\right) \leqslant H^{l+1}. \tag{7.4c}$$

This will clearly occur if $p < l+1$ for all parameter values.

We summarize, then, by noting that for

$$\frac{k(p+1)}{2} > l+1 > p, \tag{7.5}$$

Eq. (2.2) has nontrivial solutions with $n/2$ oscillations provided s is sufficiently large and $\mu > \pi^2 n^2 D_a$. Alternately, for

$$\max\left\{ p, \frac{k(p+1)}{2} \right\} < l+1, \tag{7.6}$$

Eq. (2.2) has nontrivial solutions provided $s > 0$ is sufficiently small. In other words, there are nonuniform patterns for (2.2) whenever $p < l+1$. We do not, however, know the stability characteristics or regions of attraction for all of these solutions.

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University of Arizona

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